

AMY ROMANELLO

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SUMMARY

Candidate for a PhD in Neuroscience with expertise in analyzing multidimensional datasets. Skilled in algorithm design and applied mathematics in multi-modal MRI analysis, including stochastics and information theory, fractal geometry, and machine learning. Proven record of high-impact publications, with exceptional project management, communication, and data visualization skills.

EDUCATION

Candidate for PhD in Medical Neuroscience

Oct 2020 - Present

Charité – Universitätsmedizin Berlin & Berlin School of Mind and Brain

Berlin, Germany

- Thesis on: *Temporal dynamics of resting-state brain signals: capturing the complexity of spontaneous brain activity in health and disease*

Master of Science in Medical Neuroscience

Sep 2018 - Oct 2020

Charité – Universitätsmedizin Berlin, GPA: 1.7 (DE) $\approx$ 3.3 (USA)

Berlin, Germany

- Thesis on: *Altered functional connectivity dynamics and disability severity in patients with multiple sclerosis*

Bachelor of Science in Behavioral Neuroscience

Aug 2013 - May 2018

Northeastern University, GPA: 3.9 (USA) $\approx$ 1.1 (DE)

Boston, MA

- Relevant coursework in Psychopharmacology; Biological Psychology; Brain, Behavior, & Immunity; Clinical Neuroscience; Genetics & Molecular Biology; Physics; Calculus & Diff. Eqs

WORK EXPERIENCE

Wissenschaftliche Mitarbeiterin, Cognitive Neurology Lab

Oct 2023 - Present

Department of Neurology, Charité – Universitätsmedizin Berlin

Berlin, Germany

PhD Candidate, Cognitive Neurology Lab

Oct 2020 - Present

Department of Neurology, Charité – Universitätsmedizin Berlin

Berlin, Germany

Research Intern, Psychotropic Substances Research Group

Jan 2019 - Present

Department of Psychiatry, Charité – Universitätsmedizin Berlin

Berlin, Germany

Research Intern (Master's Student), Cognitive Neurology Lab

Jul 2019 - Sep 2020

Department of Neurology, Charité – Universitätsmedizin Berlin

Berlin, Germany

Community Residence Counselor, McLean Hospital

Jul 2015 - Sep 2018

Cambridge Residence for DBT in Borderline Personality Disorder

Cambridge, MA, USA

Mental Health Counselor, North Shore Medical Center

Jul 2015 - Aug 2017

Inpatient Child and Adolescent Psychiatry Unit

Lynn, MA, USA

Undergrad. Research Assistant, <u>Center for Translational Neuroimaging</u> PI: Craig Ferris, Northeastern University Dept. of Psychology	Aug 2016 - Dec 2016 Boston, MA, USA
Undergrad. Research Assistant, <u>Aggression Lab</u> PI: Richard Melloni Jr, Northeastern University Dept. of Psychology	May 2014 - Dec 2014 Boston, MA, USA

TEACHING EXPERIENCE

MATLAB Programming Fixed-term lecturer (paid) Complete course design (WiSe 24/25)	Oct 2024 - March 2025 Berlin, Germany
Maps and Measures: Applied Neuroscience Methods Fixed-term lecturer (unpaid) 2 lectures/semester (SoSe 23, 25) <i>Functional connectivity analysis and its clinical application in multiple sclerosis</i> <i>BOLD-signal pattern congruency: explaining FC alterations in autoimmune encephalitis</i>	April 2025 - Sep 2025 April 2023 - Sep 2023 Berlin, Germany
Seminar in Cognitive Neurology Fixed-term lecturer (unpaid) 1 lectures/semester (SoSe 24, 26) <i>Functional brain dynamics and cognition</i>	April 2026 - June 2026 April 2024 - Sep 2024 Berlin, Germany
Clinical Neuroscience Tutorial 2 sessions/semester (WiSe 23/24, 24/25, 25/26)	Oct 2025 - March 2026 Oct 2024 - March 2025
All courses for Master Program: Mind & Brain Department of Psychology, Humboldt-Universität zu Berlin	Oct 2023 - March 2024 Berlin, Germany

TECHNICAL SKILLS

Listed in order of active use (updated 04/2026)

Programming languages - MATLAB (6.5 years) - Python (1.5 years) - R (5 years) - Shell scripting (4 years)	Development tools - Git, GitHub - Pycharm, VSCode - Docker - Asana, Miro	MRI tools - FreeSurfer - fMRIPrep & CONN Toolbox - FSL - SPM
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AD-HOC REVIEWER

Neuroimage: Clinical
Human Brain Mapping
Neurology

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

Organization for Human Brain Mapping , Member	May 2024 - Present
Nonviolent Crisis Intervention (CPI) Certification for Mental Healthcare Providers	2015 - 2018
Medication Administration Program (MAP) Certification for Community Residence Counselors	2016 - 2018
Emergency Medical Technician (EMT-B) Licensed with Dept. of Public Health, Massachusetts, USA	2014 - 2017

SELECTED PUBLICATIONS

Romanello, A., Krohn, S., von Schwanenflug, N., Chien, C., Bellmann-Strobl, J., Ruprecht, K., Paul, F., & Finke, C. (2022). Functional connectivity dynamics reflect disability and multi-domain clinical impairment in patients with relapsing-remitting multiple sclerosis. **NeuroImage: Clinical**, 36, 103203. <https://doi.org/10.1016/j.nicl.2022.103203>

Romanello, A., von Schwanenflug, N., Franka, M., Paul, F., Prüss, H., Krohn, S.*, Finke, C.* (2026). Functional connectivity is linked to symbolic BOLD patterns: replication, extension, and clinical application of the human 'complexome'. **In Press. Network Neuroscience**. Preprint: <https://doi.org/10.1101/2025.09.05.674447>

Krohn, S., **Romanello, A.**, von Schwanenflug, N., Rasmussen J. M., Buss, C., Valk, S. L., Madan, C. R., & Finke, C. (2026). Fractal analysis of brain shape formation predicts age and genetic similarity in human newborns. **Nature Neuroscience**. 29(1), 171–185. <https://doi.org/10.1038/s41593-025-02107-w>

Krohn, S., von Schwanenflug, N.*, Waschke, L.*, **Romanello, A.**, Gell, M., Garrett, D. D., & Finke, C. (2023). A spatiotemporal complexity architecture of human brain activity. **Science Advances**, 9(5). <https://doi.org/10.1126/sciadv.abq3851>

Krohn, S.*, Müller-Jensen, L.*, Kuchling, J., **Romanello, A.**, Wurdack, K., Rekers, S., Bartsch, T., Leyboldt, F., Paul, F., Ploner, C. J., Prüss, H., & Finke, C. (2025). Cognitive Deficits in Anti-LG1 Encephalitis Are Linked to Immunotherapy-Resistant White Matter Network Changes. **Neurology Neuroimmunology & Neuroinflammation**, 12(2), e200360. <https://doi.org/10.1212/NXI.0000000000200360>

von Schwanenflug, N., Ramirez-Mahaluf, J. P., Krohn, S., **Romanello, A.**, Heine, J., Prüss, H., Crossley, N. A., & Finke, C. (2023). Reduced resilience of brain state transitions in anti-N-methyl-D-aspartate receptor encephalitis. **European Journal of Neuroscience**, 57(3), 568–579. <https://doi.org/10.1111/ejn.15901>

Hartung, T.J., Steigerwald, F., **Romanello, A.**, Kodde, C., Endres, M., Frank, S., Heuschmann, P., Koehler, P., Krohn, S., Pape, D., Schaller, J., Stöcklein, S., Vadasz, I., Vehreschild, J., Witzenrath, M., Zoller, T., Finke, C. on behalf of the NAPKON Study Group. (2025). Post-COVID fatigue is associated with reduced cortical thickness after hospitalization. **Annals of Clinical and Translational Neurology**, 0, 1–12. <https://doi.org/10.1002/acn3.70260>

Schwichtenberg, K., Hartung, T., Heine, J., Krohn, S., Boesl, F., Rust, R., **Romanello, A.**, Paul, F., Bellmann-Strobl, J., Franke, C., Finke, C. (2026). Association of structural brain changes with cognitive deficits and fatigue in patients with post COVID-19 condition. **Brain Communications**. fcag099. <https://doi.org/10.1093/braincomms/fcag099>

ADDITIONAL PUBLICATIONS

Bendau, A., Viohl, L., Petzold, M. B., Helbig, J., Reiche, S., Marek, R., **Romanello, A.**, Moon, D. U., Gross, R. E., Masah, D. J., Gutwinski, S., Mick, I., Montag, C., Evens, R., Majić, T., & Betzler, F. (2022). No party, no drugs? Use of stimulants, dissociative drugs, and GHB/GBL during the early COVID-19 pandemic. **International Journal of Drug Policy**, 102, 103582. <https://doi.org/10.1016/j.drugpo.2022.103582>

Brandt, L., Evens, R., Reiche, S., Marek, R. M., Moon, D. U., Groß, E., **Romanello, A.**, Masah, D. J., Scicchitano, M., Gutwinski, S., Montag, C., Majić, T., & Mick, I. (2021). Predictors of Alcohol Consumption Among Younger Adults During the First Phase of the COVID-19 Pandemic. **Frontiers in Psychiatry**, 12, 748158. <https://doi.org/10.3389/fpsyt.2021.748158>

Evens, R., Reiche, S., Marek, R. M., Moon, D. U., Groß, R. E., **Romanello, A.**, Jalilzadeh Masah, D., Scicchitano Böckheler, M., Gutwinski, S., Montag, C., Mick, I., & Majić, T. (2021). Psychedelic Experiences During the Early COVID-19 Pandemic: Findings From an International Online Survey. *Frontiers in Psychiatry*, 12, 732028. <https://doi.org/10.3389/fpsy.2021.732028>

Mielau, J., Reiche, S., Moon, D. U., Groß, E., Gutwinski, S., Betzler, F., **Romanello, A.**, Masah, D. J., Scicchitano, M., Marek, R., Brandt, L., Evens, R., Mick, I. M., Majić, T., & Montag, C. (2023). Cannabis use during the early COVID-19 pandemic: Use patterns, predictors, and subjective experiences. *Frontiers in Psychiatry*, 13, 1037451. <https://doi.org/10.3389/fpsy.2022.1037451>

Mielau, J., Evens, R., Reiche, S., Marek, R., Moon, D. U., Groß, E., **Romanello, A.**, Masah, D. J., Brandt, L., Gutwinski, S., Montag, C., Majić, T., & Mick, I. M. (2024). Consumption Patterns of Benzodiazepines and Opioids Drawn from an Online Survey in the Early COVID-19 Pandemic. *SUCHT*, 70(1), 45–55. <https://doi.org/10.1024/0939-5911/a000853>

WORKING PAPERS & PREPRINTS

Romanello, A.*, Anderhalten, L*, Bahr, L., Krohn, S., Böken, O. J., Schmitz-Hübsch, T., Bellmann-Strobl, J., Mähler, A., Finke, C., & Paul, F. Longitudinal increases in blood ketone levels are linked to functional connectivity changes in multiple sclerosis. *Manuscript in preparation*.

Romanello, A., Stöckigt, B., White, L., Marek, R. M., & Majić, T. On the safety and suitability of psychedelic-assisted therapy as a treatment for cluster-B personality disorders: insights from an online survey study. *Manuscript in preparation*.

Krohn, S., Cammà, G., Zhao, W., **Romanello, A.**, Wurdack, K., Böken, O. J., Rekers, S., Paul, F., Prüß, H., & Finke, C. (2025). Memory dysfunction and psychiatric outcomes in anti-NMDA receptor encephalitis are linked to altered structural brain complexity. *Submitted. Preprint:* <https://doi.org/10.1101/2025.09.22.677264>

Zhao, W., Wurdack, K., Bartels, F., Cammà, G., Bozem, A.E., **Romanello, A.**, Mantwill, M., Prüß, H., Paul, F., Ploner, C.J., Madan, C.R., Krohn, S.* & Finke, C.* (2026). Structural brain changes are linked to clinical and cognitive outcomes in anti-LGI1 encephalitis. *Submitted*.

CONFERENCE POSTERS

Romanello, A., et al. Tissue-specific age effects of fractal dimensionality in the neonatal brain: a simulation study. To be presented at the **32nd Annual Meeting of the Organization for Human Brain Mapping**. 2026. Bordeaux, France.

Anderhalten, L.*, **Romanello, A.***, et al. Longitudinal increases in blood ketone levels are linked to functional connectivity changes in MS. Presented at the **10th Annual Americas Committee for Treatment and Research in Multiple Sclerosis (ACTRIMS) Forum**. 2025. West Palm Beach, Florida, USA.

Romanello, A., et al. Reduced functional connectivity in autoimmune encephalitis is explained by BOLD pattern incongruency. Presented at the **30th Annual Meeting of the Organization for Human Brain Mapping**. 2024. Seoul, South Korea.

Romanello, A., et al. Time-resolved BOLD-signal complexity: an out-of-sample replication study. Presented at the **29th Annual Meeting of the Organization for Human Brain Mapping**. 2023. Montreal, Canada.

Romanello, A., et al. Functional connectivity dynamics vary with disease severity in patients with multiple sclerosis. Presented at the **28th Annual Meeting of the Organization for Human Brain Mapping**. 2022. Glasgow, Scotland.

*These authors contributed equally to this work.